

Predicting Tornadoes

In Oklahoma City

By: Taylor Murray

Erdoş Fall 2025 Data Science Boot
Camp

Our Guiding Question and Data

Question: Are surface level meteorological feature successful in predicting the likelihood of a tornado occurring?

Data Sources:

- NOAA Integrated Surface Database (ISD)
 - Collected data from 9 different Stations in/surrounding Oklahoma City from 2000-2021
- NOAA National Centers for Environmental Information. Storm Events Database.
 - A collection of data on all types of storm events from 2000-2021.

Data Merging

To answer our question on tornado prediction, we merged our data sources in the following way:

We create a label called 'TORNADO_OCCURRENCE'. This label is true for a station data sample if a tornado occurred within 3 hours after the station's observation and if the tornadic event began within 75 kilometers of the station's location.

These two hyperparameters can be varied for future projects.

Modeling Approach

	Model	validation_avg_precision	validation_roc_auc	validation_brier_score
8	XGBoost_ROS	0.026379	0.868135	0.002157
9	RF_SMOTE	0.021322	0.906564	0.002157
5	XGBoost_Conservative	0.020201	0.879479	0.002157
10	XGBoost_SMOTE	0.019554	0.874139	0.002159
7	RF_ROS	0.018930	0.883368	0.002160
6	XGBoost_Aggressive	0.018468	0.852950	0.002161
3	DecisionTree_Balanced	0.017315	0.813653	0.002164
4	RandomForest_Balanced	0.016624	0.890378	0.002161
0	LogReg_L2	0.013075	0.891068	0.002172
1	LogReg_L1	0.013055	0.891176	0.002172
2	DecisionTree	0.010702	0.692706	0.002315

Train-Validation-Test Split

Train [2000-2017] → Val [2018-2019] → Test [2020-2021]

Scikit-Learn CV Calibration and GridSearchCV with Custom Splitter

Fold 1: Train [2000] → Val [2001-2002]

Fold 2: Train [2000-2003] → Val [2003-2004]

...

Fold 17: Train [2000-2015] → Val [2016-2017]

Justification: emulates realistic model deployment where we train on all historical data and predict on the next time period.

Handle Extreme Label Imbalance with SMOTE/ROS

```
===== Train Validation Split =====  
Train period: 2000-01-01 00:00:00 to 2017-12-31 22:53:00  
Test period: 2017-12-31 23:52:00 to 2019-12-31 23:52:00  
Train size: 509,364 (90.5%)  
Test size: 53,669 (9.5%)  
Train tornado rate: 0.1480%  
Test tornado rate: 0.2180%
```

Feature Engineering

Of the features that we engineered, the following helped model performance:

- a. TMP-DEW (Temperature-Dew spread)
- b. Cyclic encodings of month of year' and 'hour of day'
 - i. sin_monthofyear
 - ii. cos_monthofyear
 - iii. sin_hourofday
 - iv. cos_hourofday

	Model	validation_avg_precision	validation_roc_auc	validation_brier_score
8	XGBoost_ROS	0.026379	0.868135	0.002157
9	RF_SMOTE	0.021322	0.906564	0.002157

In addition, dropping a number of original columns helped model performance too.

	Model	Parameters	validation_Threshold	validation_avg_precision	v
0	TC_DT_RF_SMOTE	{'cv': YearlyTemporalWindowSplit(end_year=2017...	0.044158	0.064942	
1	DROP_TC_DT_RF_SMOTE	{'cv': YearlyTemporalWindowSplit(end_year=2017...	0.041303	0.071004	

Final Model

- Our final model is a tuned version of our model from the previous slide. Used GridSearchCV with custom splitter.
- Our final model trained on years 2000-2019 did not perform accurately on our held out test years (2020-2021).

	Model	train_avg_precision	TEST_avg_precision
0	final_model	0.319678	0.006182

Test vs Train Average Precision

Why:

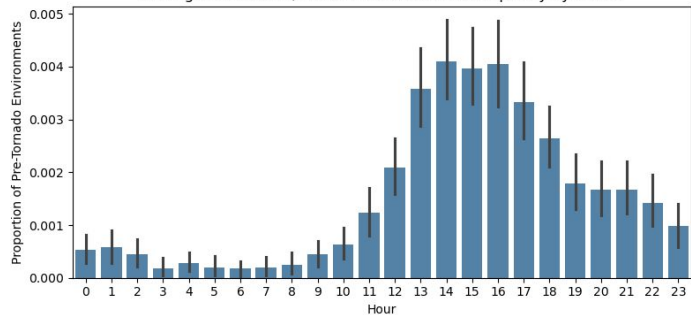
- 1) **Temporal Distribution Shift:** Atmospheric patterns shifted between training (2000-2019) and test (2020-2021) periods.
- 2) **Extreme class imbalance:** 0.15% tornado rate amplifies impact of distribution shifts
- 3) **Reliance on Seasonality:** Our model relies too heavily on what month it is.
- 4) **Limited feature set:** Surface weather observations alone provide insufficient signal. Future project may want to utilize radar data too.

TRAIN_VAL	TORNADO	MONTH	COUNTS
DATE			
May			
445			
April			
161			
October			
68			
March			
64			
June			
38			
August			
33			
November			
27			
February			
14			
December			
12			
July			
6			
January			
3			
Name: count, dtype: int64			
TEST	MONTH	TORNADO	COUNTS
DATE			
October			
37			
May			
12			
July			
9			
June			
9			
January			
6			
Name: count, dtype: int64			

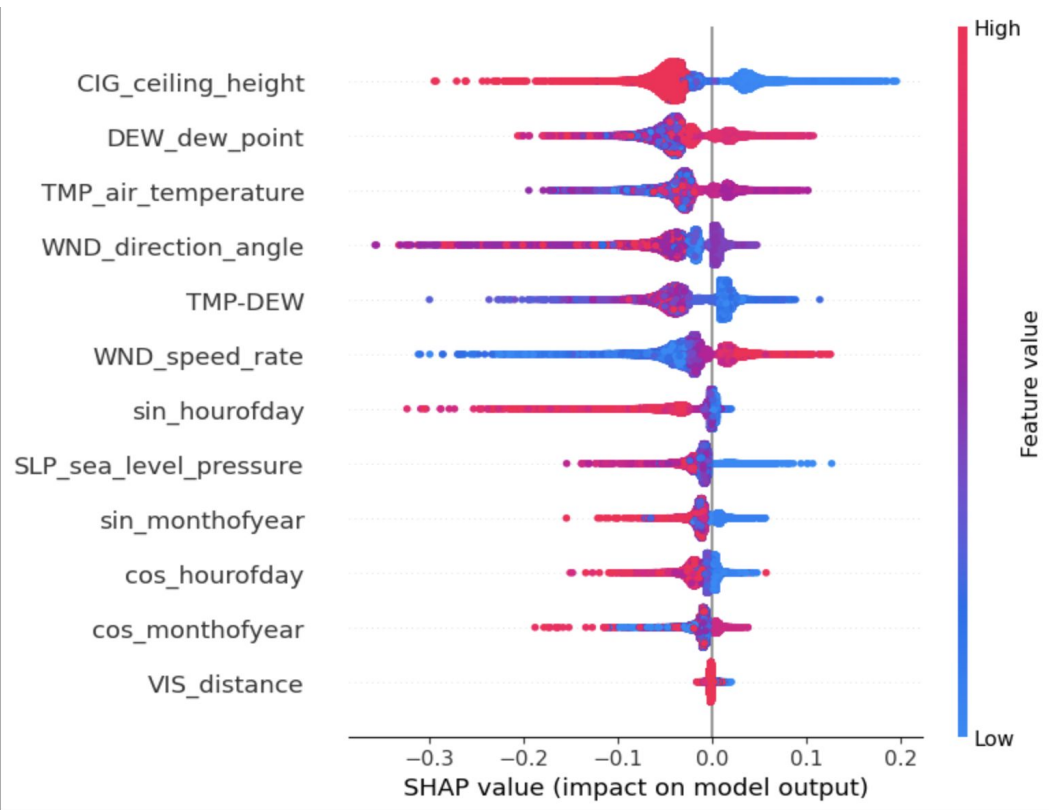
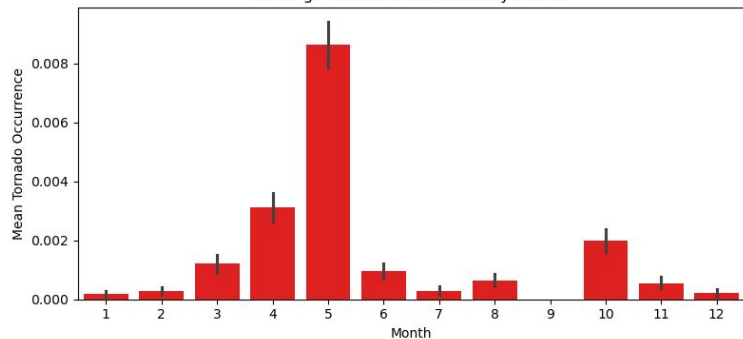
Monthly Tornado Count

Temporal Distribution + Feature Importance

Average Pre-Tornado/Tornado Environmental Frequency by Month



Average Tornado Occurrence by Month



Future Work

1. Investigate Temporal Distribution Shift Through Ensemble Methods:

The significant validation to test performance drop reveals temporal distribution shifts in tornado formation patterns. Future work could explore ensemble approaches that combine models trained on different **seasonal** periods to create more robust predictions across distributional shifts. In addition, we should also take into account the **time of day** in dealing with temporal distribution.

2. Incorporate Radar and Satellite Data

Current features are limited to surface weather station measurements (temperature, pressure, dewpoint, ect...). Integrating Doppler radar data—particularly mesocyclone detection, hook echo patterns, and reflectivity values—could dramatically improve tornado prediction accuracy.